

Midterm Project

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Project title

A Comparative Analysis of PM2.5 Levels: Los Angeles vs. New York City in 2023

Introduction

PM2.5 (fine particulate matter) is a key indicator of air quality and public health. Considering the 2023 Canadian wildfire smoke impact on the East Coast, I want to do an analysis to compare between these two counties.

The Question: Is there a significant difference in the daily PM2.5 concentrations between a West Coast hub (Los Angeles) and an East Coast hub (New York City) during the year 2023?

Data Preparation

Source: US EPA Daily Data (<https://www.epa.gov/outdoor-air-quality-data/download-daily-data>).

Data Wrangling:

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.6
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.1      v tibble     3.3.1
v lubridate  1.9.4      v tidyr      1.3.2
```

```
v purrr      1.2.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(janitor)
```

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

```
chisq.test, fisher.test
```

```
library(knitr)
library(lubridate)

# Load the data
ny_data <- read_csv("NY.csv") %>%
  mutate(across(everything(), as.character), State = "New York")
```

Rows: 943 Columns: 22

```
-- Column specification -----
Delimiter: ","
chr (10): Date, Source, Units, Local Site Name, AQS Parameter Description, M...
dbl (12): Site ID, POC, Daily Mean PM2.5 Concentration, Daily AQI Value, Dai...
```

```
i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
ca_data <- read_csv("CA.csv") %>%
  mutate(across(everything(), as.character), State = "California")
```

Rows: 4824 Columns: 22

```
-- Column specification -----
Delimiter: ","
chr (12): Date, Source, Site ID, Units, Local Site Name, AQS Parameter Descr...
dbl (10): POC, Daily Mean PM2.5 Concentration, Daily AQI Value, Daily Obs Co...
```

```
i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```

air_data_raw <- bind_rows(ny_data, ca_data) %>%
  clean_names()

air_data <- air_data_raw %>%
  mutate(
    date = mdy(date),
    daily_mean_pm2_5_concentration = as.numeric(daily_mean_pm2_5_concentration),
    site_id = as.character(site_id))

# Filter for a specific county if your file contains the whole state
air_filtered <- air_data %>%
  filter(county %in% c("New York", "Los Angeles")) %>%
  select(date, daily_mean_pm2_5_concentration, county, state) %>%
  drop_na()

# Create seasons
air_filtered <- air_filtered %>%
  mutate(
    month_num = month(date),
    season = case_when(
      month_num %in% c(12, 1, 2) ~ "Winter",
      month_num %in% c(3, 4, 5) ~ "Spring",
      month_num %in% c(6, 7, 8) ~ "Summer",
      month_num %in% c(9, 10, 11) ~ "Fall"
    ),
    season = factor(season, levels = c("Winter", "Spring", "Summer", "Fall"))
  ) %>%
  select(-month_num)

```

Exploratory Analysis

Dataset Characteristics:

```

# Calculate the values
total_obs <- nrow(air_filtered)
start_date <- min(air_filtered$date)
end_date <- max(air_filtered$date)

# Print them using cat()

```

```
cat("Dataset Characteristics:\n",
    "-----\n",
    "Total Observations :", total_obs, "\n",
    "Start Date           :", as.character(start_date), "\n",
    "End Date              :", as.character(end_date), "\n")
```

Dataset Characteristics:

```
-----
Total Observations : 5767
Start Date         : 2023-01-01
End Date           : 2023-12-31
```

Variables included, distribution of key variables.

```
# 1. Print the variables included in the dataset
cat("Variables included in the analysis:\n")
```

Variables included in the analysis:

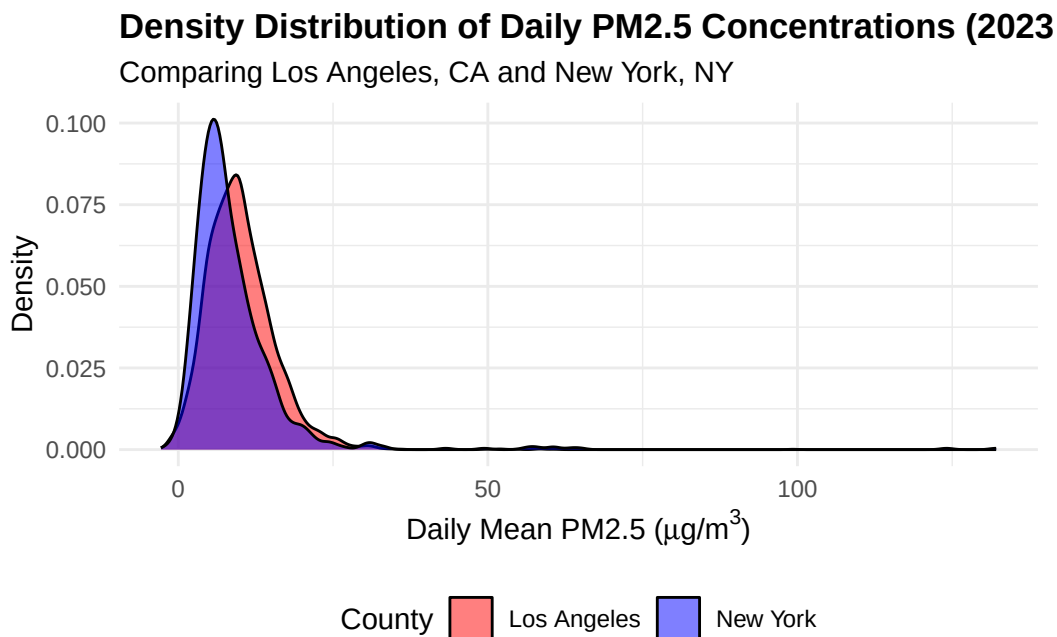
```
cat(paste(names(air_filtered), collapse = ", "), "\n\n")
```

date, daily_mean_pm2_5_concentration, county, state, season

```
# 2. Plot the distribution of the key variable (PM2.5)
library(ggplot2)

ggplot(air_filtered, aes(x = daily_mean_pm2_5_concentration, fill = county)) +
  geom_density(alpha = 0.5) +
  theme_minimal() +
  labs(
    title = "Density Distribution of Daily PM2.5 Concentrations (2023)",
    subtitle = "Comparing Los Angeles, CA and New York, NY",
    x = expression(paste("Daily Mean PM2.5 (", mu, "g/", m^3, ")")),
    y = "Density",
    fill = "County"
  ) +
  scale_fill_manual(values = c("Los Angeles" = "red", "New York" = "blue")) +
```

```
theme(
  plot.title = element_text(face = "bold"),
  legend.position = "bottom"
)
```



Preliminary Results

Figure 1: Time-Series Plot. This figure helps identify spikes, like wildfire events.

```
ggplot(air_filtered, aes(x = date, y = daily_mean_pm2_5_concentration, color = county)) +
  geom_line(alpha = 0.7) +
  geom_hline(yintercept = 35, linetype = "dashed", color = "darkred", linewidth = 0.8) +
  annotate("text", x = min(air_filtered$date), y = 38,
    label = "EPA NAAQS Limit (35 µg/m³)", color = "darkred", hjust = 0, size = 3.5) +
  theme_minimal() +
  labs(
    title = "Daily PM2.5 Trends Throughout 2023",
    subtitle = "Spikes above the dashed red line indicate unhealthy air quality days",
    x = "Date",
    y = expression(paste("Daily Mean PM2.5 (", mu, "g/", m^3, ")")),
    color = "County"
```

```
) +
scale_color_manual(values = c("Los Angeles" = "red", "New York" = "blue")) +
theme(legend.position = "bottom", plot.title = element_text(face = "bold"))
```

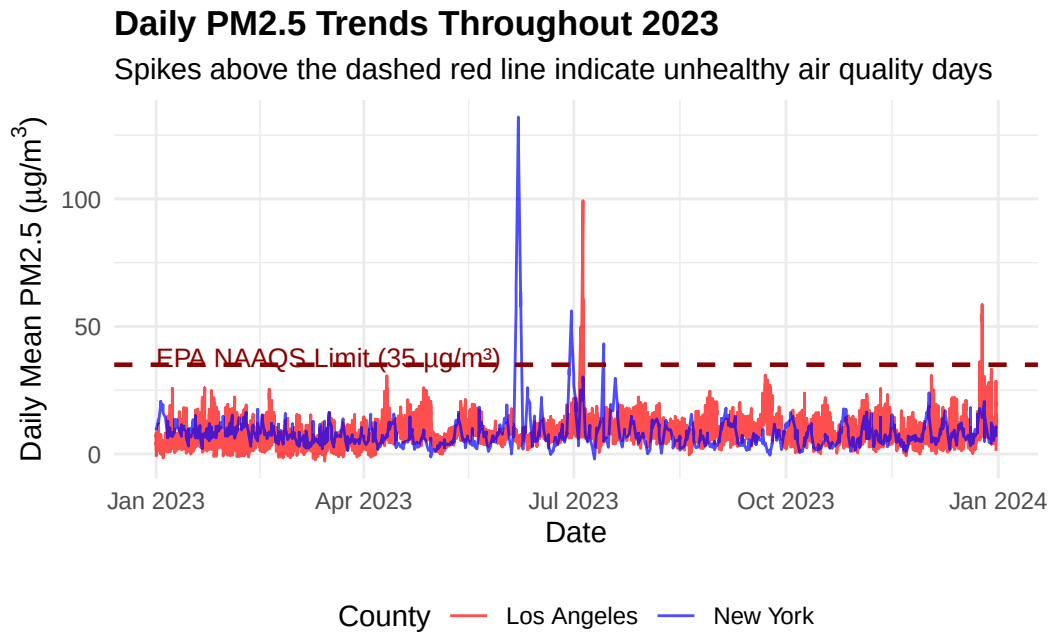


Figure 2: Seasonal Boxplot. A boxplot comparing PM2.5 levels by Season and City. This shows if one city is consistently worse in the Summer vs. Winter.

```
ggplot(air_filtered, aes(x = season, y = daily_mean_pm2_5_concentration, fill = county)) +
geom_boxplot(alpha = 0.7, outlier.alpha = 0.5, outlier.shape = 21) +
theme_minimal() +
labs(
  title = "Seasonal Distribution of PM2.5 (2023)",
  subtitle = "Comparing seasonal variations between Los Angeles and New York",
  x = "Season",
  y = expression(paste("PM2.5 Concentration (", mu, "g/", m^3, ")")),
  fill = "County"
) +
scale_fill_manual(values = c("Los Angeles" = "red", "New York" = "blue")) +
theme(legend.position = "bottom", plot.title = element_text(face = "bold"))
```

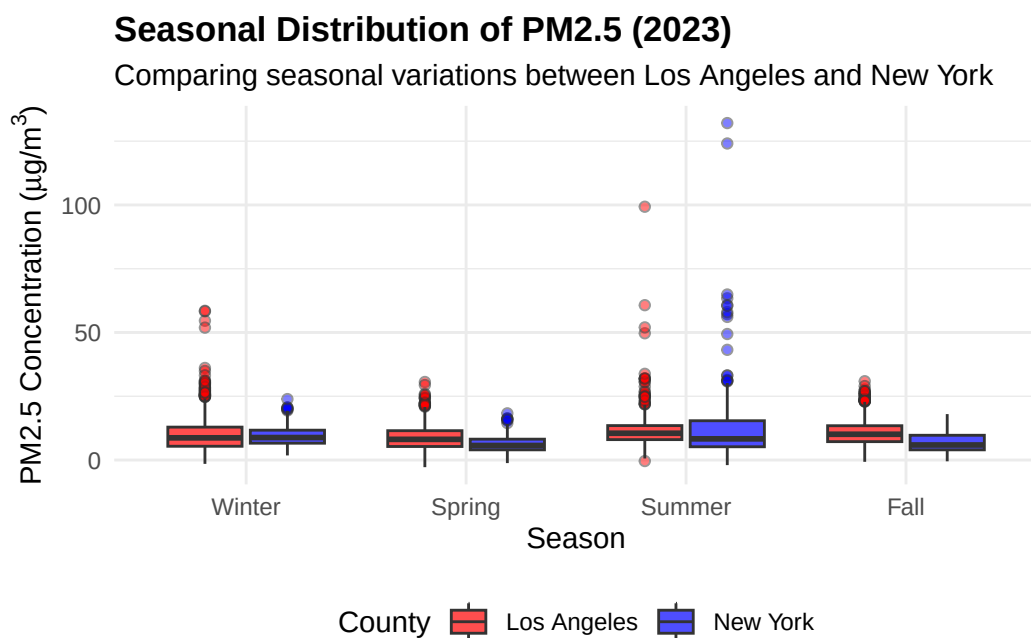


Table 1: Summary statistics for PM 2.5

```
summary_stats <- air_filtered %>%
  group_by(county, state) %>%
  summarise(
    Mean_PM25 = mean(daily_mean_pm2_5_concentration),
    Median_PM25 = median(daily_mean_pm2_5_concentration),
    Max_PM25 = max(daily_mean_pm2_5_concentration),
    SD_PM25 = sd(daily_mean_pm2_5_concentration),
    .groups = "drop"
  )

kable(summary_stats,
  col.names = c("County", "State", "Mean", "Median", "Max", "Std Dev"),
  caption = "Table 1: Descriptive Statistics of Daily PM2.5 in 2023",
  digits = 2)
```

Table 1: Table 1: Descriptive Statistics of Daily PM2.5 in 2023

County	State	Mean	Median	Max	Std Dev
Los Angeles	California	10.08	9.5	99.3	5.66
New York	New York	8.87	7.0	132.1	8.90

Table 2: A kable table showing the number of days each city exceeded the EPA National Ambient Air Quality Standard (NAAQS) for PM2.5 (35 $\mu\text{g}/\text{m}^3$).

```
# Calculate days exceeding the EPA limit
exceedance_table <- air_filtered %>%
  group_by(county, State = state) %>%
  summarise(
    Total_Days_Recorded = n(),
    # sum() with a logical condition counts how many times it was TRUE
    Days_Exceeding_NAAQS = sum(daily_mean_pm2_5_concentration > 35, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(Percent_Unhealthy = round((Days_Exceeding_NAAQS / Total_Days_Recorded) * 100, 1)) %>%
  arrange(desc(Days_Exceeding_NAAQS))

kable(exceedance_table,
      col.names = c("County", "State", "Total Days", "Days > 35  $\mu\text{g}/\text{m}^3$ ", "% Unhealthy Days"),
      caption = "Table 2: Number of Days Exceeding EPA NAAQS for PM2.5 in 2023",
      align = "l1ccc")
```

Table 2: Table 2: Number of Days Exceeding EPA NAAQS for PM2.5 in 2023

County	State	Total Days	Days > 35 $\mu\text{g}/\text{m}^3$	% Unhealthy Days
New York	New York	943	11	1.2
Los Angeles	California	4824	9	0.2

Conclusion

Summary New York County experienced cleaner air than Los Angeles County. As shown in the summary statistics, New York had a noticeably lower median PM2.5 concentration (7.0 $\mu\text{g}/\text{m}^3$) compared to Los Angeles (9.5 $\mu\text{g}/\text{m}^3$). Furthermore, Los Angeles showed consistently higher baseline pollution across most seasons, particularly in the Fall and Winter.

Key Finding The key finding of this analysis is that New York's air quality was far more volatile. As visualized in the time-series plot and seasonal boxplots, New York experienced a massive spike in PM2.5 levels during the summer—specifically in early June. During this event, PM2.5 levels skyrocketed to a maximum of 132.1 $\mu\text{g}/\text{m}^3$, which heavily skewed New York's summer distribution. This severe anomaly aligns with the historic Canadian wildfire smoke that blanketed the East Coast in June 2023.

Because of these extreme summer spikes, New York actually recorded a higher proportion of unhealthy air quality days exceeding the EPA's 35 $\mu\text{g}/\text{m}^3$ limit (1.2%) compared to Los Angeles (0.2%), despite Los Angeles having a higher daily average.

Finally, while Los Angeles suffers from a consistently higher baseline of fine particulate matter, New York is susceptible to far more severe, acute pollution events.